Akoizo Original Practice - Form 1 - Answer key

Division B | Thermodynamics | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) A sealed conducting container is necessarily…

Unit: Systems, temperature, and thermal equilibrium.
Worked-case data: Convert 27 °C to Kelvin and compare a 10 °C temperature change.

A. Closed, not necessarily isolated

B. Always isolated

C. Always open to matter

D. Without internal energy

Correct: A

2. (2 points) Triple and critical points are identical?

Unit: Phases, diagrams, and transition paths.
Worked-case data: During ideal melting at 0 °C, added energy increases what?

A. No

B. Always

C. Only for water

D. Only at zero pressure

Correct: A

3. (2 points) Ideal monatomic Cv equals…

Unit: Ideal gases and kinetic molecular theory.
Worked-case data: For one mole at 300 K in 0.010 m^3, estimate pressure using R=8.314 J/(mol K).

A. 5R

B. Zero

C. 3R/2

D. R/2

Correct: C

4. (2 points) Heat and work are…

Unit: Internal energy and the first law.
Worked-case data: A system receives 500 J heat and does 200 J work. Find delta U and chemistry work.

A. Always state functions

B. Always equal

C. Always positive

D. Path-dependent

Correct: D

5. (2 points) Carnot efficiency uses…

Unit: Heat engines, Carnot cycles, refrigerators, and heat pumps.
Worked-case data: For Th=600 K and Tc=300 K, find Carnot efficiency and refrigerator COP.

A. Kelvin reservoir temperatures

B. Celsius directly

C. Only pressure

D. Only gas mass

Correct: A

6. (2 points) Heat is…

Unit: Systems, temperature, and thermal equilibrium.
Worked-case data: Convert 27 °C to Kelvin and compare a 10 °C temperature change.

A. Identical to temperature

B. Only mechanical work

C. Boundary energy transfer

D. A stored substance

Correct: C

7. (2 points) At a melting plateau, phase fraction can…

Unit: Phases, diagrams, and transition paths.
Worked-case data: During ideal melting at 0 °C, added energy increases what?

A. Only become plasma

B. Only decrease under heating

C. Change

D. Never change

Correct: C

8. (2 points) Gas-law pressure should be…

Unit: Ideal gases and kinetic molecular theory.
Worked-case data: For one mole at 300 K in 0.010 m^3, estimate pressure using R=8.314 J/(mol K).

A. Absolute

B. Gauge without conversion

C. Always atmospheric

D. Only a color

Correct: A

9. (2 points) A complete cycle has net delta U…

Unit: Internal energy and the first law.
Worked-case data: A system receives 500 J heat and does 200 J work. Find delta U and chemistry work.

A. Equal to temperature

B. Zero

C. Always positive

D. Always negative

Correct: B

10. (2 points) Equal-temperature molecules all have one speed?

Unit: Historical experiments and modern thermodynamic reasoning.
Worked-case data: A mechanical heater adds 1000 J to a system with total capacity 500 J/K, neglecting losses. Predict delta T.

A. Only at 300 K

B. No

C. Always

D. Only for monatomic gases

Correct: B

11. (2 points) Convection coefficient h is…

Unit: Conduction, convection, radiation, and heat capacity.
Worked-case data: Heat 0.20 kg water by 10 K with c=4180 J/(kg K), neglecting losses.

A. Universal for all air

B. Equal to mass

C. Always zero

D. Condition-dependent

Correct: D

12. (2 points) Before solving final temperature with ice, check…

Unit: Latent heat, heating curves, and calorimetry.
Worked-case data: Melt 0.10 kg ice at 0 °C using Lf=334 kJ/kg. Find energy.

A. Whether all ice melts

B. Only container color

C. Only elapsed time

D. Only the answer units

Correct: A

13. (4 points) Explain “Isothermal” in your own words.

Unit: Gas processes with graphs and simple algebra.
Worked-case data: Pressure: 100,000 Pa, held constant. Volume change: An increase of 0.002 m^3. Heat input: 300 J enters the gas.
Separate investigation observations:
Pressure: 100,000 Pa, held constant.
Volume change: An increase of 0.002 m^3.
Heat input: 300 J enters the gas.

answer: Constant temperature. (4 points).

14. (4 points) Explain “System” in your own words.

Unit: Energy direction and the second law.
Worked-case data: Heat input: An engine receives 500 J. Useful work: It delivers 150 J. Cycle: The working substance returns to its initial state.
Separate investigation observations:
Heat input: An engine receives 500 J.
Useful work: It delivers 150 J.
Cycle: The working substance returns to its initial state.

answer: The part of the world chosen for study. (4 points).

15. (2 points) A model’s historical importance makes it universally exact?

Unit: Historical experiments and modern thermodynamic reasoning.
Worked-case data: A mechanical heater adds 1000 J to a system with total capacity 500 J/K, neglecting losses. Predict delta T.

A. Always

B. Only if famous

C. Only if old

D. No

Correct: D

16. (4 points) Why use Kelvin in radiation?

Unit: Conduction, convection, radiation, and heat capacity.
Worked-case data: Heat 0.20 kg water by 10 K with c=4180 J/(kg K), neglecting losses.

answer: Absolute fourth-power temperatures determine the model. (4 points).

17. (4 points) List three stages from cold ice to warm water.

Unit: Latent heat, heating curves, and calorimetry.
Worked-case data: Melt 0.10 kg ice at 0 °C using Lf=334 kJ/kg. Find energy.

answer: Warm ice, melt ice, warm liquid. (4 points).

18. (6 points) Solve this case and explain your reasoning: Pressure: 100,000 Pa, held constant. Volume change: An increase of 0.002 m^3. Heat input: 300 J enters the gas.

reasoning: Calculate work: 100,000 x 0.002 = 200 J.
Use change in internal energy = heat added - work done.
Subtract 300 - 200 = 100 J. (3 points).

conclusion: An increase of 100 J (3 points).